

Kowshik S

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EDUCATION

Anna University — College of Engineering, Guindy (CEG) <i>Bachelor of Engineering in Computer Science and Engineering — CGPA: 7.73/10</i>	Chennai, Tamil Nadu 2023 – 2027 (Expected)
Vidyaa Vikas Matric Higher Secondary School <i>Class XII, Matriculation Board — 97.5% (585/600)</i>	Coimbatore, Tamil Nadu 2022 – 2023

EXPERIENCE

Department of Computer Science & Engineering, Anna University, CEG <i>Web Development Intern — JWT-Based Login System for Student-Faculty Portal</i>	Chennai, Tamil Nadu May 2026 – Jun 2026
- Engineered a role-based authentication system (separate student/faculty login flows) using JWT and bcrypt on a full-stack MERN application. - Implemented device fingerprinting to validate that each API request originates from the same device/browser as the original login, preventing a copied or stolen token from being reused elsewhere.	

PROJECTS

PhotoShare — Photo Sharing Platform (Live · Repo)	TrizenAI — Full-Stack Challenge
- Developed a full-stack event photo platform solo (React 18, Node/Express, PostgreSQL via Prisma) with role-based access control for Admin and Team Member roles, enforced at the API and UI layers. - Designed two-scope JWT auth — long-lived session tokens plus short-lived scoped tokens issued after PIN verification — streaming all photo access through the API with no public storage URLs exposed. - Wrote 30 automated tests (Vitest/Supertest) covering auth and gallery flows, and deployed across 3 cloud services (Vercel, Render, Supabase), resolving a production IPv6-only DB connectivity issue via pooling.	
URL Shortener (Live · Repo)	Jul 2026
- Shipped a full-stack URL shortener enforcing per-user link ownership through JWT-based stateless auth and bcrypt password hashing. - Designed a Redis cache-aside layer in front of PostgreSQL for link redirects and per-user/IP rate limiting, and logged click analytics (count, referrer, user-agent, timestamp) asynchronously via FastAPI BackgroundTasks. - Delivered the React/TypeScript frontend with Tailwind (authenticated dashboard, analytics visualizations, QR code generation, dark mode), and set up CI/CD with GitHub Actions to deploy the Dockerized app to Render.	
Title-Based Playlist Recommendation System with Bias Mitigation	Dec 2025 – Apr 2026
- Developed an ML pipeline generating playlists from titles via Sentence-BERT embeddings, K-Means clustering, and a transformer fine-tuned with Cross-Entropy and Triplet Loss over the Spotify Million Playlist Dataset. - Built a voting-based recommendation mechanism with LLM-assisted re-ranking to reduce popularity bias, evaluating across 58,584 playlists at a mean NDCG@66 of 0.4210 (benchmark range: 0.4156–0.4311).	

TECHNICAL SKILLS

Languages	C/C++, Python, Java, JavaScript, TypeScript, SQL, HTML/CSS
Frameworks & Databases	React.js, Node.js, Express.js, FastAPI, Tailwind CSS, MongoDB, PostgreSQL, Redis
Tools & DevOps	Git, GitHub, Docker, GitHub Actions (CI/CD)
Concepts	REST APIs, JWT, bcrypt
Problem Solving	Solved 300+ DSA problems on LeetCode, HackerRank, and GeeksforGeeks
Machine Learning	PyTorch, Scikit-learn, Sentence Transformers, HuggingFace Transformers, Pandas, NumPy, Recommender Systems
Coursework	Data Structures and Algorithms, Operating Systems, Computer Networks, Database Management Systems, Machine Learning

ACHIEVEMENTS & EXTRACURRICULAR ACTIVITIES

Python for Data Science — NPTEL, secured a Top 5% rank in the nationwide certification exam (2025)	
Anna University — College of Engineering, Guindy (CEG) <i>National Service Scheme (NSS) — Joint Secretary</i>	Chennai, Tamil Nadu Jul 2025 – Jun 2026
Organized a village outreach camp; led two campus initiatives — "Inaippu" (voter ID registration) and "Mellinam" (career guidance).	